

SWARNABHA DUTTA

Full Stack Developer (MERN) — React, Node.js, MongoDB, TypeScript — AI Integration — Automated Deployment
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SUMMARY

I build backend systems that are fast, secure, and production-ready — then I make them faster. As a Full Stack Engineer specializing in **MERN + TypeScript + AI integration**, I've shipped systems that cut API response times by **93%** (19s → 1.3s) and improved cached response speed by **667%** through intelligent Redis caching strategies. Experienced in integrating LLM APIs, building secure multi-role authentication systems, and delivering end-to-end fullstack features on production infrastructure.

TECHNICAL SKILLS

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Redux
Backend: Node.js, Express.js, REST APIs, Python, JWT Authentication
Databases: MongoDB, PostgreSQL, MySQL, Prisma ORM
Caching: Redis (Upstash)
Auth & Security: JWT, RBAC, Clerk
Tools & Deployment: Git, GitHub, Docker, Render, Vercel, GitHub Actions, CI/CD
AI Integration: LLM APIs (Google Gemini), Prompt Engineering
Problem Solving: 284+ LeetCode problems solved (DSA + Algorithms), Java

PROJECTS

- AI Code Reviewer** | Full Stack AI-Powered Code Analysis Platform | [GitHub](#) | [Live](#) Mar 2025 – Jun 2025
- Built a production-ready **TypeScript-based MERN application** leveraging the **Google Gemini API** to analyze uploaded or pasted source code for bugs, security vulnerabilities, anti-patterns, and performance optimization opportunities.
 - Optimized frontend performance through rendering improvements, asset optimization, and TypeScript migration, increasing the **Lighthouse Performance score from 82 to 98** while improving maintainability and scalability.
 - Implemented **Redis intelligent caching**, reducing backend API latency by **93% (19s → 1.3s)** and improving repeated response speed by **667% (20s → 30ms)**.
 - Reduced redundant **Gemini API requests** and **MongoDB queries** by **82.4%** through cache-hit optimization, significantly lowering infrastructure cost and backend workload.
 - Built a production-grade testing suite with **137+ automated tests** achieving over **84% line coverage** using Vitest, React Testing Library, MSW, accessibility testing, and integration testing.
 - Developed secure REST APIs with JWT authentication, modular Express architecture, centralized error handling, and automated production deployment using Render GitHub Deploy Hooks.
- HealthCheckBuddy** | Full Stack Healthcare Platform | [GitHub](#) | [Live](#) Apr 2025 – May 2025
- Built full stack telemedicine platform using **Next.js, PostgreSQL, Prisma ORM** with role-based workflows for Patients, Doctors, and Admins — including doctor verification, appointment booking, and credit management.
 - Implemented **atomic database transactions** for credit deduction, payout processing, and appointment cancellations — ensuring data consistency across concurrent operations.
 - Integrated **Vonage Video API** for real-time doctor-patient video consultations with secure token generation and session management.
 - Enforced **five-layer security architecture** including Clerk authentication, RBAC, server-side token validation, session expiry handling, and input sanitization.

INTERNSHIP EXPERIENCE

- Prodigy Infotech** Nov 2024 – Dec 2024
Frontend Web Development Intern Remote
- Delivered **4 production-grade responsive web applications** within weekly sprint cycles, practicing version control workflows and deadline-driven delivery in a professional codebase environment.
 - Implemented UI/UX features including smooth scrolling, hover animations, and dynamic content rendering using vanilla JavaScript and DOM manipulation.

EDUCATION

- Guru Nanak Institute of Technology** Kolkata, West Bengal
B.Tech in Electronics and Communication Engineering Aug 2021 – Jun 2025
- CGPA: **8.33/10**